

Member ID: _____

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SERVER ADMINISTRATION USING MICROSOFT (310)

NATIONAL 2023

Total Points (50 @ 2 points each) _____ **(100 points)**

Test Time: 60 minutes

GENERAL GUIDELINES:

Failure to adhere to any of the following rules will result in disqualification:

1. Member must hand in this test booklet and all printouts if any. Failure to do so will result in disqualification.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.

Multiple Choice Questions

Directions: Identify the letter of the choice that best completes the statement or answers the question.

1. Your company has purchased another company that also uses Windows Server 2012 R2 and Active Directory. Both companies need to be able to access each other's forest resources. How can you achieve this goal with the least administrative effort?
 - A. Configure selective authentication
 - B. Configure an external trust
 - C. Create a two-way forest trust
 - D. Share the global catalog for both companies
2. You need to make a DNS resource record for an IPv6 host address. Which of the following would be the correct resource record type?
 - A. A record
 - B. AAAA record
 - C. IP6 record
 - D. CNAME record
3. What application pools is created by default when you install the Web Server role?
 - A. DefaultAppPool
 - B. WebAppPool
 - C. IIS-Pool
 - D. DefaultWebPool
4. What component of Kerberos is responsible for storing keys for encrypting and decrypting data in the authentication process?
 - A. Ticket Granting Server (TGS)
 - B. Key Management Host (KMS)
 - C. Key Distribution Center (KDC)
 - D. Kerberos Authentication Gateway (KAG)
5. Which type of IPV6 configuration uses router advertisements for prefix assignment?
 - A. Static autoconfiguration
 - B. Stateless autoconfiguration
 - C. EUI-64 autoconfiguration
 - D. Stateful autoconfiguration
6. When selecting authentication types for your RAS server, what is considered to be the most security authentication type?
 - A. Password Authentication Protocol (PAP)
 - B. Extensible Authentication Protocol (EAP)
 - C. Challenge Handshake Authentication Protocol (CHAP)
 - D. Challenge Handshake Authentication Protocol version 2 (CHAPv2)

7. .com, .net, and .edu are all examples of what level of the DNS name hierarchy?
 - A. Stub domain
 - B. Subdomain
 - C. Top-level domain names
 - D. Second-level domain names
8. Timestamps within Kerberos are used to help guard against a _____.
 - A. Spoofing attack
 - B. Replay attack
 - C. TCP SYN attack
 - D. DDOS attack
9. You have recently added a Linux server to handle a new application. You want to integrate user authentication between Linux and your existing Windows Server 20012 R2 domain controllers. What can you do to complete this task?
 - A. Create an external trust
 - B. Create a transitive trust
 - C. Create a forest trust
 - D. Create a realm trust
10. Which service provided by PKI ensures that a party cannot dispute the validity of the transaction?
 - A. Authentication
 - B. Nonrepudiation
 - C. Confidentiality
 - D. Integrity
11. You need to validate DNS server responses. Which of the following is a suite of features and protocols for doing this?
 - A. DNSSEC
 - B. EUI-64
 - C. DNS Proxy
 - D. Kerberos
12. Which authentication standard is utilized for Windows Server 2016 if Active Directory is not installed?
 - A. Kerberos
 - B. Windows NT Lan Manager version 1
 - C. Windows NT Lan Manager version 2
 - D. Windows Advanced Authentication Manager
13. The standard DACL for a package object assigns read permissions to what group by default?
 - A. Domain Admins
 - B. Authenticated Users
 - C. Everyone
 - D. Domain Users

14. You are assigning user roles to users within your Active Directory environment for use with VMM. What role should you assign a user that needs to manage self-service users as well as create and manage their own virtual machines?
- A. Delegated administrator
 - B. Fabric administrator
 - C. Tenant administrator
 - D. Self service manager
15. What happens if you shut down a clustered Windows Server 2012 R2 Hyper-V server without pausing it first?
- A. The VMs are suspended until the server is restarted
 - B. The VMs are automatically live migrated
 - C. The VMs are removed from the cluster
 - D. The VMs are shut down and exported to another server
16. By default, how long are deleted objects stored in the Active Directory database before they are removed entirely?
- A. 60 days
 - B. 120 days
 - C. 160 days
 - D. 180 days
17. You would like to design your AD DS to support business-to-business relationships where the account federation server validates credentials and no Active Directory trust is created. Which of the following AD DS designs should you choose?
- A. Web SSO
 - B. Federated Web SSO with forest trust
 - C. AD RMS Federated trust
 - D. Federated Web SSO
18. After experiencing several issues with an Active Directory domain controller, you have decided to perform a restore operation on Active Directory. What Advanced Boot Option should be used to perform the restoration?
- A. Safe Mode with Networking
 - B. Directory Services Repair Mode
 - C. Directory Restoration Mode
 - D. Active Directory Emergency Maintenance Mode
19. You need to configure a failover cluster and want a quorum configuration in which the cluster quorum data is located in a shared folder on a server in another location. Which of the follow quorum configurations would you need?
- A. Node Majority
 - B. Node and Disk Majority
 - C. Node and File Share Majority
 - D. No Majority: Disk Only

20. Which statement is NOT a true statement to keep in mind when designing an OU structure in Active Directory?
- A. Nested OUs should be used sparingly to maintain simplified administration.
 - B. Create of OUs involves more processing resources because each request through an OU requires CPU time.
 - C. Active Directory works more efficiently when OUs are set up horizontally.
 - D. Microsoft recommends nesting OUs at least four levels deep.
21. In DNS, what type of zone contains a read/write master copy of all resource records for the zone?
- A. Primary zone
 - B. Secondary zone
 - C. Stub zone
 - D. Global zone
22. You need to test the functionality of DNS operations on a domain controller, as well as troubleshoot issues with DNS forwarders, delegation, and record registration. Which tool would be best to complete this task?
- A. Using DNS Manager
 - B. Using dcdiag /test:dns
 - C. Using dnscmd.exe
 - D. Using nslookup
23. A customer owns a computer without a TPM chip but wants to implement BitLocker Drive Encryption. Which of the following would you recommend to the customer?
- A. The computer will require the use of a smart card reader that can be used to authenticate the user.
 - B. The computer will require a password on a cold start up to ensure that the drive contents are accessed by authorized users only.
 - C. The computer will utilize BitLocker encryption, but security will be provided at the software level as opposed to the hardware level.
 - D. The computer will require a USP flash drive that contains a personal identification number used to start the computer and secure the drive.
24. Selecting the “Allow the connection if it is secure” option when creating a Windows firewall rule relies on what encryption protocol default?
- A. AES
 - B. IPSec
 - C. MD5
 - D. SHA-1

25. You have a technician that needs to be able to backup files and directories, restore files and directories, and shut down the system. Without giving the technician broad administrative access to the domain, what group should you assign the technician?
- A. Power Users group
 - B. Enterprise Admins group
 - C. Domain Admins group
 - D. Server Operators group
26. What type of DNS query causes a DNS server to respond with the information it currently has in its local database?
- A. Iterative query
 - B. Extended query
 - C. Local query
 - D. Recursive query
27. What type of DNS resource record is used to identify a server that is authoritative for a zone?
- A. SOA record
 - B. NS record
 - C. PTR record
 - D. MX record
28. You need to setup folder redirection setting to redirect everyone's folders to the same location. What option would perform this task?
- A. Advanced
 - B. Targeted
 - C. Basic
 - D. Open
29. What type of DNS server response includes a list of name servers?
- A. Referral
 - B. Recursive
 - C. Iterative
 - D. Resolver
30. By default, replication between domain controllers, when no changes have occurred, is scheduled to happen at what interval?
- A. Never
 - B. Once per hour
 - C. Once per day
 - D. Once per week

31. Which of the following encryption standards is utilized with Kerberos on Windows Server 2016 and Windows 10?
- A. Blowfish
 - B. AES
 - C. 3DES
 - D. DES
32. Which of the following is true regarding the domain functional level?
- A. All domain controllers and member servers must be running the Windows version that supports the functional level
 - B. You must raise the functional level on all domain controllers
 - C. You can have different functional levels within the forest
 - D. The domain and forest functional level must be the same
33. If a domain consists of domain controllers that are running versions of Windows server earlier than Windows Server 2008, what replication method is used?
- A. Downlevel Server Replication (DSR)
 - B. Distributed File System Replication (DFSR)
 - C. File Replication Service (FRS)
 - D. NT Replication Service (NTRS)
34. What authentication protocol can perform authentication, but does not require it, so that operating systems without password encryption capabilities can still connect to RAS?
- A. Challenge Handshake Authentication Protocol
 - B. Extensible Authentication Protocol
 - C. Challenge Handshake Authentication Protocol version 2
 - D. Password Authentication Protocol
35. How does a DHCP server respond to an IP request that can't be renewed?
- A. The server sends a DHCPNACK
 - B. The server sends a DHCPACK
 - C. The server does not respond
 - D. The server forwards the request
36. In Event Viewer, what statement regarding the ability to filter events is **NOT** true?
- A. Events can be filters based on logged time.
 - B. Events can be filtered by IP address for network events.
 - C. Events can be filtered based on level.
 - D. Events can be filtered based on the user associated with the event.
37. You need to install a domain controller on which changes can't be written. What is the name for a domain controller that can perform this?
- A. Read only domain controller
 - B. Access only domain controller
 - C. No write domain controller
 - D. Secured domain controller

38. What PowerShell cmdlet will install the Active Directory Domain Services role?
- A. Add-WindowsFeature ActiveDirectory
 - B. Add-WindowsFeature AD-Services
 - C. Add-WindowsServices AD-Domain-Services
 - D. Add-WindowsFeature AD-Domain-Services
39. You are investigating strange traffic on your network and wish to resolve an IP address to DNS name. What resource record is used to do a reverse lookup?
- A. SOA record
 - B. AAAA record
 - C. PTR record
 - D. MX record
40. What command line utility can be used to display the default route, as well as all routing table entries?
- A. ipconfig
 - B. route
 - C. show route
 - D. net print route
41. What type of Certificate Authority (CA) in the three-level hierarchy is sometimes referred to as a policy CA and issues certificates to issuing CAs?
- A. Intermediate
 - B. Enterprise
 - C. Root
 - D. Offline
42. You have a network of 1000 users. You discover that you must frequently restore files from backup that users accidentally deleted or older versions of the file is required. What feature can enable you to reduce the amount of restoration procedures you must perform?
- A. AD Sysvol
 - B. Shadow copies
 - C. File classification
 - D. DFSR
43. What Active Directory partition holds the DNS database?
- A. Domain directory partition
 - B. Configuration partition
 - C. Application directory partition
 - D. Global catalog partition
44. The IP address 129.53.82.10 falls under what class of IP addresses?
- A. A
 - B. B
 - C. C
 - D. D

45. You have several thousand documents in the Accounting share. You need to be able to easily identify those files that contain sensitive data. Which of the following features can help with this task?
- A. Storage Reports
 - B. Dynamic Access Control
 - C. File classification
 - D. BranchCache
46. What Windows feature is designed to prevent malware, such as computer viruses, Trojan horses, and worms, by preventing programs from using system memory space?
- A. Data Execution Prevention
 - B. Protected Memory Blocking
 - C. System Memory Guard
 - D. Data Memory Protection
47. You are setting up a Windows Deployment Server (WDS). Which of the following options in DHCP do you need to configure so client computers can contact the WDS?
- A. 060 PXEClient
 - B. 080 WDS config
 - C. 046 WINS/NBT Node type
 - D. 044 WINS/NBNS Servers
48. An employee leaves your company, what should happen to any certificates held by that employee that were issued by the company's PKI?
- A. The should be forwarded to the AIA
 - B. The should be put on the CRL
 - C. They should be included in the CDP
 - D. They should be added to the OR
49. When using Rights Management Services (RMS) security, documents are encrypted using an AES key and what other additional key?
- A. DSA
 - B. RSA
 - C. ECDSA
 - D. ECRSA
50. What Kerberos account policy can be configured to encourage users sign out after a certain amount of time?
- A. Enforce user logon restrictions
 - B. Maximum lifetime for user ticket renewal
 - C. Maximum lifetime for user ticket
 - D. Maximum lifetime for service ticket